
Atomic Thinking of LLMs: Decoupling and Exploring Mathematical Reasoning Abilities

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Abstract

Large Language Models (LLMs) have demonstrated outstanding performance in mathematical reasoning capabilities. However, we argue that current large-scale reasoning models primarily rely on scaling up training datasets with diverse mathematical problems and long thinking chains, which raises questions about whether LLMs genuinely acquire mathematical concepts and reasoning principles or merely remember the training data. In contrast, humans tend to break down complex problems into multiple fundamental atomic capabilities. Inspired by this, we propose a new paradigm for evaluating mathematical atomic capabilities. Our work categorizes atomic abilities into two dimensions: (1) **field-specific abilities across four major mathematical fields**, algebra, geometry, analysis, and topology, and (2) **logical abilities at different levels**, including conceptual understanding, forward multi-step reasoning with formal math language, and counterexample-driven backward reasoning. We propose corresponding training and evaluation datasets for each atomic capability unit, and conduct extensive experiments about how different atomic capabilities influence others, to explore the strategies to elicit the required specific atomic capability. Evaluation and experimental results on advanced models show many interesting discoveries and inspirations about the different performances of models on various atomic capabilities and the interactions between atomic capabilities. Our findings highlight the importance of decoupling mathematical intelligence into atomic components, providing new insights into model cognition and guiding the development of training strategies toward a more efficient, transferable, and cognitively grounded paradigm of “*atomic thinking*”.

1 Introduction

In recent years, as Large Language Models (LLMs) have achieved remarkable performance in language understanding [1–7], visual perception [8–15], complex reasoning [16–21] agentic intelligence [22–25] and honesty [26–29], mathematical reasoning has emerged as a key focus for LLM cognitive abilities [30–33]. Mathematics, as a fundamental reasoning task, offers both verifiable answers and a wide range of difficulty levels, drawing increasing research attention [34–36]. Recent studies, particularly those involving reasoning models such as OpenAI’s o1 and DeepSeek-R1, have demonstrated strong mathematical performance, achieving impressive results on many challenging benchmarks [37–39].

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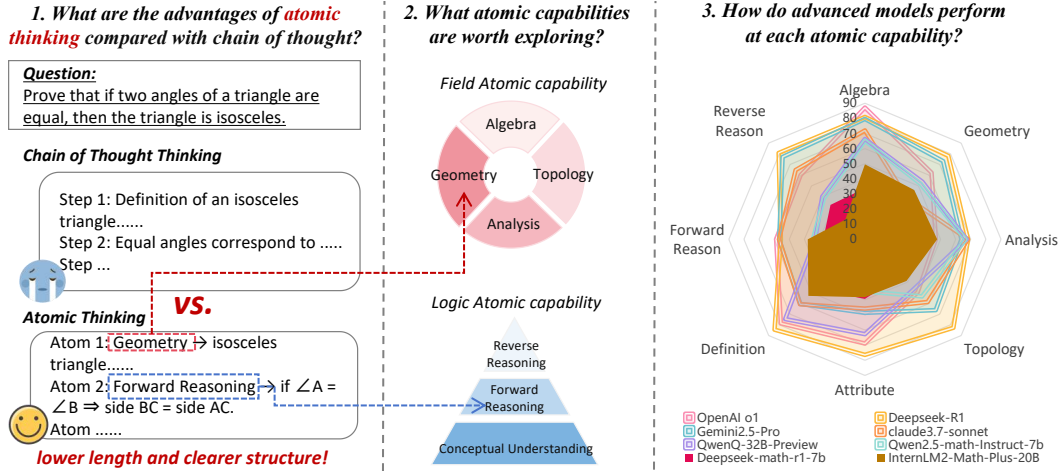


Figure 1: This figure illustrates an overview of our atomic thinking. It compares thought chains and atomic thinking, highlighting the efficiency of atomic thinking. Next, it shows the atomic capabilities we focus on. Finally, it provides the performance of advanced models in every atomic capability.

Current approaches enhance mathematical performance by scaling up training data, incorporating diverse mathematical problems, and complex reasoning paradigms [40, 41, 31]. WizardMath [32] synthesizes complex multi-field data via varied math instructions, while models like OpenAI o1 train extended chains of thought to perform higher-level reasoning. As the complexity of mathematical problems increases, some studies, such as Lean4 [42] and Lean-STaR [43], leverage formal mathematical languages to mitigate ambiguity in natural language multi-step reasoning, as well as enhancing the reward model for problem-solving process [44–46]. However, as model size and data scale approach saturation, these training paradigms are facing challenges:

- *Do models truly grasp mathematical concepts and inference patterns, or are they merely memory problems by chain of thought reasoning training?*
- *Is there a more fundamental cognitive atom that can break through the current paradigm, and what are the advantages of this atomic thinking compared with the chain of thought?*

Figure 1 compares the differences between the two paradigms. Existing math reasoning strategies, such as chain-of-thought and tree-of-thought, tend to rely heavily on sequential context [34, 47–49]. This not only leads to inefficient use of computational resources but also introduces noise through excessive self-correction in long reasoning chains. In contrast, human reasoning typically decomposes complex problems into atomic problems, solving them incrementally and integrating only the essential information for subsequent steps, which is referred to as *atomic thinking* [50]. This atomic thinking paradigm promotes flexible and structural reasoning, enabling more efficient problem-solving like data probe [51–53]. Thus, decoupling the mathematical atomic abilities of LLMs is not only essential for assessing their cognitive depth but also key to transitioning from the current “question drilling” paradigm to a “*atomic thinking*” framework. So we wonder *what atomic capabilities are worth exploring?*

Current mathematical benchmarks mostly assess models’ accuracy in end-to-end problem solving [54–56], offering little insight into a systematic assessment of atomic capabilities. To address this, we propose a novel framework for exploring mathematical atomic abilities, encompassing both field and logical reasoning capabilities, with an emphasis on ensuring minimal overlap between different atomic units while maintaining broad coverage of mathematical tasks. For field atomic abilities, we draw inspiration from modern mathematics and construct four foundational fields: **algebra, geometry, analysis, and topology**. For logical reasoning abilities, we reference cognitive psychology in math reasoning to define three core capabilities: (1) **Conceptual Understanding**, which grasps math definitions and axioms; (2) **Forward Reasoning with Formal Math Language**, which conducts rigorous, multi-step reasoning using symbolic systems; (3) **Counterexample-driven Backward Reasoning**, requiring constructing counter-examples and leveraging backward reasoning. For each atomic ability, we construct training and testing data, ensuring interpretability and isolating cross-

ability. We conduct the evaluation experiments as shown in Figure 1 to explore: *how do advanced models perform at each atomic capability?*

Beyond decoupled atomic capability evaluation, we also conduct composite experiments across reasoning levels and varied atomic abilities to investigate how atomic abilities influence each other. Our findings on all the results reveal several key insights:

- **Field-level performance:** LLMs perform better in algebra and analysis, while struggling in geometry and topology. Interestingly, models exhibit atypical behavior in topology, performing worse on easier tasks yet better on harder ones.
- **Logical reasoning abilities:** Larger LLMs exhibit stronger conceptual understanding, likely due to superior pretraining memory. However, even advanced commercial models struggle with constructing counterexamples, indicating a gap in backward reasoning skills.
- **Cross-field interaction:** Training on low-difficulty data can hinder high-level skill expression in some fields. Notably, activating algebraic abilities significantly improves performance in other fields, which is often more than direct training in the target field.
- **Cross-Logic interaction:** Conceptual understanding enhances other reasoning abilities and field atomic abilities. Surprisingly, training solely on definition completion tasks suffices to stimulate high-level ability, outperforming models trained on more complex data. This reveals the supporting value of conceptual understanding in mathematical training.

2 Related work

Mathematics-enhanced large language models Mathematical reasoning has become a key focus in exploring the upper bounds of LLMs’ cognitive capabilities [57, 58]. Unlike general-purpose models such as GPT-4 [37] and Gemini [30], math-enhanced LLMs emphasize field-specific strategies, including data augmentation, pretraining, fine-tuning [59], and reinforcement learning on large-scale mathematical corpora [36]. **WizardMath** [32] synthesizes diverse math data through instruction generation and leverages RLHF and process supervision. **NuminaMath** [60] adopts Tool-Integrated Reasoning (TIR) to generate math data, including questions with fine-grained solutions. **Qwen2.5-Math** [31] fine-tunes Qwen2.5 on proprietary high-quality math data. **InternLM2-Math** [35] enhances logical rigor by integrating formal mathematical language, code interpreters, and theorem proving in Lean4. **Deepseek-Math-rl** [34] focuses on data engineering and efficient RL training.

Mathematical benchmarking Due to their objective correctness and structured difficulty [61, 62], math tasks are ideal for evaluating LLMs [54, 63, 64, 55, 56]. Benchmarks like **MATH** [65] and **GSM8K** [66] test high school and elementary-level reasoning and have become standard. To meet the demands of advanced models, more challenging datasets such as **OlympiadBench** [67] target competition-level exams. Formal theorem proving benchmarks like [68] **Putnam Bench**, **CoqGym** [63], and **MiniF2F** [64] further assess logical reasoning with tools like Coq and Lean. However, most benchmarks assess end-to-end question-solving performance without decomposing tasks into **atomic reasoning abilities**. Our work aims to decouple and analyze these atomic abilities and their interactions to support finer-grained reasoning evaluation and lightweight but effective training.

3 Atomic capability decoupling and interaction

Inspired by the atomization of human cognition, we propose decoupling the mathematical capabilities of LLMs into atomic abilities. We categorize atomic capabilities into two major types, ensuring that each capability is disentangled from the others while jointly covering a wide range of mathematical tasks. We construct corresponding training and test sets to evaluate the model performance of different atomic abilities. Beyond evaluating individual capabilities, we further investigate their interactions. Specifically, we explore how stimulating one atomic capability may affect others, offering insights into how such interactions can be leveraged to enhance targeted atomic abilities and promote compositional problem-solving strategies.

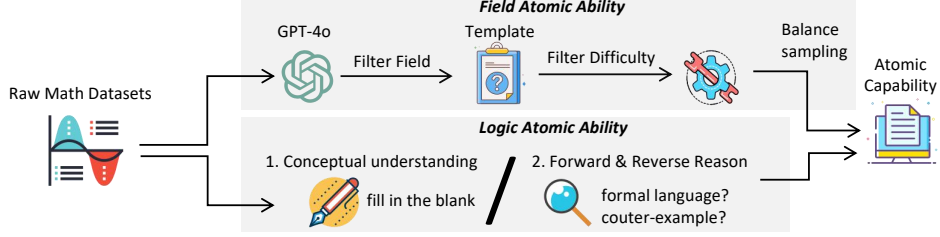


Figure 2: This figure illustrates our data construction procedure.

3.1 Atomic capability design

Field atomic capabilities We refer to the core field division of modern mathematics. It is worth noting that if we divide the capabilities too finely (e.g., dividing them into more than ten types), the atomic units will be too loose and the interaction between atomic abilities will be too complex, which may not be a particularly significant correlation. Therefore, we finally divide the field’s atomic abilities into **Algebra**, **Geometry**, **Analysis**, and **Topology**. To analyze the influence of difficulty levels in a more fine-grained aspect, we further divide each field into low difficulty (level 1) and high difficulty (level 2). Similarly, we divide it into two difficulty levels to avoid complex interactions.

Logical reasoning atomic capabilities For logic reasoning atomic ability, we refer to the human cognitive mode, deconstruct the general process of complex reasoning, and identify core abilities:

1. **Conceptual understanding** is a fundamental, which includes *definition identification* and *attribute description*. The definition identification task requires the LLM to complete the name of the corresponding definition in a statement. The property description task requires completing the detailed description in a mathematical definition, including the premises, the conditions, the key words, and parameters.
2. **Forward reasoning with formal language**: With reference to human cognitive processes, we designate forward multi-step reasoning as a higher-level logical atomic ability. Since natural language reasoning is often confronted with vague proofs and uncritical assumptions, we emphasize the forward reasoning using formal mathematical language.
3. **Backward reasoning with counterexamples**: In addition to step-by-step forward multi-step reasoning, backward reasoning with counterexamples is also a very important ability in mathematical reasoning. By skillfully constructing appropriate counterexamples, proofs can be effectively accomplished that are difficult to perform directly with forward reasoning.

It is worth noting that we do not consider computational abilities. This is because we are mainly concerned with the logical reasoning ability of mathematical reasoning. Computational capability is difficult to decouple from the reasoning process. In addition, when computations are needed, LLM invoking relevant mature computational tools would be more efficient and accurate.

3.2 Data construction for capability evaluation

To decouple each atomic capability, we construct the training and testing data, which contributes to evaluation and further exploration of the interaction. The overview of our data construction is shown in Figure 2. More data statistics and examples can be found in Appendix A.

For field capabilities, we collect data from current benchmarks such as MATH [65], GSM8K [66], Gaokao-Bench [69], OlympiadBench [67], AIME, MMLU [70], and DeepMath [71]. Using a combination of template matching and LLM-assisted annotation, we reclassify problems into four fields. When there are original field labels in the raw data, LLM takes more account of the primitive labels to better align with human thinking. Difficulty annotation is done by template matching. We take into account the data sources of the original questions (e.g., primary and secondary school questions or Olympiad questions), the original difficulty labeling information, and we emphasize aligning the difficulty classifications of different fields. After that, we randomly sample the data to ensure that the number of each field is relatively balanced. Finally, we randomly divide the training set and test set with a ratio of 3:1. For **conceptual understanding**, we extract math definitions and axioms from NaturalProofs [72] and generate fill-in-the-blank questions. For **forward reasoning**, we

collect questions and proofs with formal language such as LeanWorkbook, and filter the questions with definite answers. For **backward reasoning**, we use a counterexample-driven reasoning statement from CounterMath [73]. Given data scarcity, we maintain a 1:1 train-test split for logic atomic capability to ensure evaluation robustness.

3.3 Atomic capability interaction

In addition to decoupling individual atomic capabilities, we also try to explore the correlation and interaction between atomic capabilities. One of the most basic strategies is trying to stimulate a certain atomic capability and observe whether it has an impact on other atomic capabilities. This will provide insight for subsequent research on how to specifically stimulate the required atomic capabilities. Therefore, we explore the following atomic capability interactions:

- **Cross-difficulty Interaction:** We investigate how training on Level 1 or Level 2 tasks within a field affects performance at the other level.
- **Cross-field Interaction:** We train on tasks from one field and evaluate transfer effects to others, particularly among related or complementary fields.
- **Logical Capability Interactions:** We examine whether conceptual understanding can support high-level ability, improving forward or backward reasoning, and whether forward and backward reasoning mutually reinforce each other. Additionally, we assess whether high-level reasoning enhances foundational understanding.
- **Reasoning-to-field Interaction:** Given the abstract nature of some fields (e.g., topology and analysis), we test whether improving logical reasoning capabilities can enhance performance in field-specific tasks, especially those requiring conceptual abstraction.

4 Experimental settings

4.1 Decoupled atomic capability evaluation

Baselines We evaluate a diverse set of LLMs with a focus on mathematical reasoning. Open-source models include **Deepseek-Math-7B-RL** [34], **Eurus-2-7B-PRIME** [74], **Qwen2.5-Math-7B-Instruct** [31], **NuminaMath-7B-TIR** [60], **InternLM2-Plus-7B/20B** [35], **Abel-7B/13B** [36], **WizardMath-7B** [32], **Mathstral-7B**³, **MetaMath-Mistral-7B** [40], **Xwin-Math-7B/13B** [41], **QwQ-32B**⁴. For *proprietary models*, we use **GPT-4o**, **OpenAI o1**⁵, **Deepseek-R1**⁶, **Claude3.7-sonnet**⁷, and **Gemini2.5-pro**⁸. This selection spans varied training paradigms, data sources, architectures, and origins (academic vs industrial). Open-source evaluations are conducted on 4× L20 48GB GPUs, while proprietary models are accessed via official APIs.

Prompts and metrics We adopt default *Chain-of-Thought* (CoT) prompting, instructing models to enclose answers in `\boxed{}` for extraction. Accuracy is computed via exact match with reference answers. For **Conceptual Understanding**, models complete missing definitions. **Forward Reasoning** requires multi-step derivations using formal mathematical language. Accuracy is used for both. For **Backward Reasoning**, we refer to previous work CounterMATH [73]: (1) F-1 score on the statement judgement, and (2) Example, Strict, Loose metric to evaluate whether the model constructs valid counterexamples. Detailed description of our prompts and metrics can be found in Appendix B.

4.2 Training for atomic capability interaction

To examine interactions among atomic abilities, we fine-tune Qwen2.5-Math-Instruct-7B using supervised LoRA training [75] on 4×L20 48GB GPUs, with a learning rate of 1.0e-5. Training about

³<https://mistral.ai/news/mathstral/>

⁴<https://qwenlm.github.io/blog/qwq-32b-preview/>

⁵<https://cdn.openai.com/o1-system-card-20241205.pdf>

⁶<https://api-docs.deepseek.com/news/news250120>

⁷<https://www.anthropic.com/claude/sonnet>

⁸<https://deepmind.google/technologies/gemini/pro/>

Table 1: Model performance on field atomic capabilities. We **bold** the optimal and underline the suboptimal of models. The low and high difficulty levels correspond to level 1 and level 2, respectively.

Field		Algebra		Geometry		Analysis		Topology	
Difficulty Level		Low	High	Low	High	Low	High	Low	High
<i>Open-source models</i>									
Model $\leq$ 7B	InternLM2-math-plus-7b	49.2	35.9	33.0	31.4	41.9	41.5	27.2	37.0
	Deepseek-math-rl-7b	52.0	33.8	35.5	<u>39.3</u>	44.6	37.6	22.6	33.7
	Eurus-2-7B-PRIME	50.7	33.9	36.5	32.3	45.4	32.0	<u>36.5</u>	23.3
	NuminaMath-7B-TIR	52.4	28.2	39.9	32.1	<u>46.8</u>	33.6	26.4	25.1
	MetaMath-Mistral-7B	<u>59.6</u>	41.4	42.4	37.7	45.2	<u>37.9</u>	20.6	25.8
	Mathstral-7B-v1.0	42.3	33.3	36.4	28.1	32.9	29.8	24.1	<u>37.1</u>
	Abel-7B-002	46.7	32.9	36.1	35.2	32.9	26.3	25.6	31.5
	Xwin-Math-7B	52.6	37.3	38.3	35.1	45.3	32.9	19.7	22.5
Model > 7B	Qwen2.5-math-Instruct-7b	80.5	65.2	57.3	51.9	67.7	66.5	52.1	53.4
	Abel-13B	63.4	47.3	42.7	41.8	40.4	35.6	30.2	24.6
	Xwin-Math-13B	78.3	51.6	<u>49.5</u>	<u>46.7</u>	<u>55.7</u>	<u>54.8</u>	<u>46.7</u>	38.3
	IntwenLM2-Math-Plus-20B	67.8	<u>49.6</u>	49.3	46.1	54.3	47.8	45.4	<u>38.9</u>
<i>Commercial models</i>									
	OpenAI o1	93.6	87.1	66.3	62.3	59.3	46.9	52.9	49.4
	GPT-4o	69.3	50.5	48.5	36.3	55.1	53.2	53.0	49.4
	Deepseek-rl	<u>83.6</u>	<u>80.9</u>	76.8	78.0	<u>70.3</u>	68.2	77.7	82.7
	Claude3.7-sonnet	83.5	72.0	58.0	39.0	80.3	<u>68.1</u>	56.0	58.6
	Gemini2.5-pro	80.5	79.6	<u>70.2</u>	<u>73.1</u>	67.5	63.8	<u>68.1</u>	<u>66.2</u>

different interactions adheres to a unified hardware setup and consistent hyperparameters. After training, evaluation follows Section 4.1 to ensure comparability across settings.

5 Analysis and discussion

5.1 Experimental analysis of field atomic capabilities

We evaluate several advanced models by assessing their performance across decoupled atomic abilities in distinct mathematical fields. The detailed results are presented in Table 1. We observe that:

Larger models exhibit stronger atomic capabilities Model performance varies significantly across scales. In general, larger models perform better, benefiting from greater capacity and more extensive training data. Among 7B-scale models, **Qwen2.5-math-Instruct** achieves notably superior results across all evaluated fields, even outperforming some larger models such as **InternLM2-Math-Plus-20B**. Analysis of its outputs shows that it generates longer reasoning chains, facilitating deeper logical inference and enhancing its problem-solving capabilities.

Algebra and analysis perform better Models tend to exhibit stronger mathematical atomic abilities in Algebra and Analysis, while performance in Geometry and especially Topology remains weaker. Since we analyze the training data of the open-sourced models we have evaluated, the results reveal that Geometry and Topology are significantly underrepresented in the training data. Moreover, although geometric problems are textually presented, they often require spatial or visual reasoning—an area where LLMs typically struggle. Consequently, current models demonstrate limited atomic capabilities in these fields. Improving performance in underrepresented areas, particularly Topology and Geometry, is a pressing research challenge. For example, the low performance on topology-related tasks may reflect a lack of understanding of abstract mathematical structures. A promising direction is to integrate core mathematical concepts during training to stimulate relevant atomic skills. We explore such cross-ability interactions in Section 5.3.

Table 2: Model performance on logic atomic capabilities, where Attr. and Def. are short names of the Attribute description and definition task. We **bold** the optimal and underline the suboptimal results.

		Concept		Forward Rea.	Backward Rea.			
		Attr. (Acc.)	Def. (Acc.)	Acc.	F-1	Example(%)	Strict(%)	Loose(%)
<i>Open-source models</i>								
Model $\leq 7B$	InternLM2-math-plus-7b	43.2	46.2	23.7	33.9	36.6	9.0	9.5
	Deepseek-math-rl-7b	<u>39.4</u>	<u>46.5</u>	27.6	32.2	65.9	18.9	20.6
	Eurus-2-7B-PRIME	23.1	27.9	41.4	<u>37.5</u>	64.8	<u>28.5</u>	<u>32.0</u>
	NuminaMath-7B-TIR	22.8	27.3	30.2	30.4	54.1	13.0	13.7
	MetaMath-Mistral-7B	19.6	25.6	28.6	31.0	26.5	0.4	0.7
	Mathstral-7B-v1.0	21.7	29.8	32.9	28.2	38.9	7.5	7.9
	Abel-7B-002	20.9	31.0	33.7	34.4	<u>66.1</u>	16.0	17.9
	Xwin-Math-7B	18.8	26.3	26.4	28.1	31.3	1.2	1.7
Model $> 7B$	Qwen2.5-math-Instruct-7b	34.4	50.3	<u>34.4</u>	38.3	74.2	30.2	33.2
	Abel-13B	31.2	48.0	37.2	22.4	24.4	0.8	0.8
	Xwin-Math-13B	29.8	45.9	33.1	<u>30.2</u>	<u>31.3</u>	1.2	1.7
	InternLM2-Math-Plus-20B	<u>38.3</u>	<u>52.9</u>	<u>37.8</u>	18.4	28.8	8.4	<u>9.5</u>
<i>Commercial models</i>								
	OpenAI o1	<u>68.9</u>	<u>78.6</u>	58.7	60.1	55.8	39.8	40.9
	GPT-4o	38.1	48.3	37.7	59.0	44.7	19.7	21.3
	Deepseek-r1	76.4	84.5	55.6	80.7	<u>86.8</u>	<u>54.2</u>	<u>65.3</u>
	Claude3.7-sonnet	45.8	60.7	<u>56.8</u>	64.8	78.0	45.0	52.5
	Gemini2.5-pro	48.6	60.4	56.7	<u>77.0</u>	90.8	65.1	75.7

Internal difficulty-level analysis We further examine model performance across different difficulty levels within each field. In Algebra, models often experience sharp drops in accuracy on high-difficulty problems, indicating a gap in advanced atomic capabilities. In contrast, performance degradation in Geometry and Analysis is less severe. This suggests the need for training paradigms that better balance basic and advanced skill acquisition to ensure robust generalization. An interesting anomaly arises in Topology, where models sometimes perform better on harder problems than on easier ones. We hypothesize that this is due to mismatches between the models’ training distributions and our evaluation data: some high-difficulty problems may incidentally align with abstract patterns the models have implicitly learned. This counterintuitive result shows that what humans find hard may not be hard for LLMs, which encourages deeper exploration into the field atomic ability decomposition.

Further results in Appendix C.2 show that training data difficulty significantly affects model performance. Notably, excessively low-difficulty training data may degrade accuracy across difficulty levels. Thus, balancing training data difficulty is essential for fostering generalizable atomic capabilities.

5.2 Experimental analysis of mathematical logical reasoning atomic capabilities

We evaluate several state-of-the-art mathematical models across multiple dimensions of logical reasoning. Detailed results are shown in Table 2. From these results, there are some insights that:

Models recognize definitions but struggle with deeper conceptual understanding All models perform better at recognizing definitions than completing missing properties, indicating a surface-level grasp of mathematical rigor. This suggests that while models can identify known concepts, they often lack precise internal representations. Larger commercial models significantly outperform smaller open-source ones; for example, deepseek-r1 scores 84.5 in definition recognition and 76.4 in property completion. This disparity reflects the importance of pretraining, where larger models benefit from superior long-range memory and MoE (mixture-of-experts) mechanisms that mitigate knowledge forgetting. Current math models emphasize problem-solving over conceptual understanding, contributing to the gap between basic concept recognition and advanced reasoning or proof tasks. Section 5.4 provides further case studies.

Structured reasoning with formal language remains a challenge for smaller models Since large-scale commercial models show better performance with the best accuracy of 58.7, the models with smaller parameters struggle with both understanding the questions with formal math language and applying formal mathematical language to reason, despite performing reasonably well in natural-language-based reasoning. This suggests that intensive “problem-drilling” may promote pattern

Table 3: Performance comparison after stimulating various *field* atomic capabilities. We color the positive↑ / negative↓ influence as green / red .

Field	Algebra		Analysis		Geometry		Topology	
Difficulty Level	Low	High	Low	High	Low	High	Low	High
Qwen-base	80.5	65.2	67.7	66.5	52.1	53.4	52.1	53.4
Qwen-train-Algebra	80.2 (↓0.3)	69.7 (↑4.5)	75.8 (↑8.1)	71.5 (↑5.0)	65.7 (↑13.6)	57.5 (↑4.1)	56.0 (↑3.9)	62.3 (↑8.9)
Qwen-train-Analysis	81.3 (↑0.8)	66.4 (↑1.2)	71.8 (↑4.1)	64.9 (↓1.6)	58.2 (↑6.1)	55.8 (↑2.4)	46.1 (↓6.0)	54.1 (↑0.7)
Qwen-train-Geometry	79.6 (↓0.9)	68.1 (↑2.9)	69.8 (↑2.1)	59.5 (↓7.0)	57.3 (↑5.2)	56.0 (↑2.6)	52.8 (↑0.7)	60.9 (↑7.5)
Qwen-train-Topology	79.7 (↓0.8)	64.4 (↓0.8)	71.5 (↑3.8)	59.2 (↓7.3)	61.7 (↑9.6)	55.6 (↑2.2)	53.7 (↑1.6)	59.1 (↑5.7)

Table 4: Performance comparison after stimulating various *field* atomic capabilities that are trained on InternLM2-math-plus-7B.

Field	Algebra		Analysis		Geometry		Topology	
Difficulty Level	Low	High	Low	High	Low	High	Low	High
InternLM2-math-plus-7B	49.2	35.9	33.0	31.4	41.9	41.5	27.2	37.0
InternLM2-train-Algebra	48.1 (↓1.1)	37.8 (↑1.9)	35.8 (↑2.8)	33.6 (↑2.2)	43.1 (↑1.2)	42.1 (↑0.6)	27.8 (↑0.6)	37.9 (↑0.9)
InternLM2-train-Geometry	48.6 (↓0.6)	36.3 (↑0.4)	32.2 (↓0.8)	30.5 (↓0.8)	44.3 (↑2.4)	44.0 (↑2.5)	28.7 (↑1.5)	38.8 (↑1.8)

memorization over structured formal reasoning. Some models attempt to bridge this gap by generating code-like representations to aid multi-step deduction. Notably, reasoning-oriented models such as o1 and DeepSeek-R1, equipped with stronger long-range inference and self-reflection, achieve outstanding results in this category.

Limited counterexample abilities reveal the limits of problem-solving training For the ability to judge the truth value of mathematical statements, open-source models average around 30 F-1 points, and even advanced models like QWQ-32B reach only 39.9, while Deepseek-R1, optimized for mathematical reasoning, scores 80.7. In generating counterexamples, the Qwen series performs particularly well, sometimes surpassing commercial models like o1. Conversely, models like MetaMath succeed in only 26.5% of such tasks. Although the best model Gemini2.5-pro demonstrates superior performance across various metrics when it constructs counter-examples, the performance of the other models on example consistency is still relatively low, with almost none exceeding 50%. This reflects the limitations of training paradigms overly focused on direct problem-solving, which hinders higher-level abstraction and conceptual reasoning.

5.3 Experimental analysis of interactions between atomic abilities

5.3.1 Influence between atomic abilities across different fields

We investigate the influence **between fields** among field atomic abilities, as summarized in Table 3. In particular, atomic abilities in *algebra* consistently exhibit **positive effects** across all other fields, with particularly strong gains observed in *Analysis* and *Geometry*. In some cases, Algebra training even yields greater improvements than in-field training. For instance, when evaluating *Geometry* atomic abilities, in-field training improved performance by 6.1 and 2.4 points at Levels 1 and 2, respectively, whereas activating Algebraic abilities led to larger gains of 13.6 and 4.2 points. This suggests a **complementary relationship** among atomic abilities, likely because algebraic problem-solving emphasizes fundamental reasoning skills that underpin more abstract fields, and our case study in Section 5.4 demonstrates that. These findings highlight the potential of leveraging **diverse, field-specific atomic abilities** to enhance target capabilities more effectively. However, we also observe **negative transfer** effects. Strengthening atomic abilities in *Topology* led to performance declines in *Algebra* and *Analysis*, with the largest drops reaching 0.8 and 7.3 points on Level 2. This may be due to substantial **data distribution divergence**. We have provided a visualization result in Appendix E. To further validate our conclusions, we have now included InternLM2-math-plus-7B as an additional baseline for training and evaluation, aiming to further strengthen our claims. The results, shown in Table 4, are consistent with our observations on Qwen2.5-math-instruct and provide a strong complement to our previous findings. These findings underscore the need to consider that,

Table 5: Performance comparison after stimulating various *logic* atom capabilities.

Atom Capability	Concept		Formal language	Counter example
	Attribute	Definition	Acc.	F-1
Qwen-base	34.4	50.3	34.4	30.2
Qwen-train-Concept	34.8 (↑0.4)	53.7 (↑3.4)	53.5 (↑19.1)	40.1 (↑9.9)
Qwen-train-Backward	30.3 (↓4.1)	46.3 (↓4.0)	50.3 (↑15.9)	41.1 (↑10.9)
Qwen-train-Forward	25.1 (↓9.3)	44.4 (↓5.9)	53.7 (↑19.3)	40.2 (↑10.0)

Table 6: Performance comparison after stimulating conceptual understanding of atomic capabilities.

Field	Algebra		Analysis		Geometry		Topology	
	Low	High	Low	High	Low	High	Low	High
Qwen-base	80.5	65.2	67.7	66.5	52.1	53.4	52.1	53.4
Qwen-train-Concept	81.3 (↑0.8)	66.1 (↑0.9)	69.7 (↑2.0)	72.3 (↑5.8)	62.3 (↑10.2)	59.8 (↑6.4)	55.6 (↑3.5)	57.2 (↑3.8)

when trying to stimulate one, interactions between fields can significantly impact performance in unintended ways.

5.3.2 Interactions among logical reasoning atomic abilities

We also explore the interaction among **logical reasoning atomic abilities** as shown in Table 5. Our findings indicate that *conceptual understanding* plays a fundamental role in supporting two higher-level reasoning abilities. Training solely on fill-in-the-blank tasks that activate conceptual comprehension atomic ability results in substantial improvements of 19.1 and 9.9 in forward and backward reasoning. This aligns with our earlier analysis: current mathematical models, especially those with **limited parameter capacity**, exhibit significant deficiencies in conceptual understanding, which constrain the model’s capacity to develop more advanced reasoning skills. Furthermore, forward reasoning and counterexample construction appear to **mutually reinforce** each other, indicating the potential for bidirectional enhancement between forward and backward reasoning atomic abilities. However, enhancing high-level atomic abilities alone can lead to a decline in conceptual understanding, revealing that **overemphasis on solving complex problems** may lead to **forgetting of basic mathematical concepts**.

5.3.3 Influence of conceptual understanding on field-specific atomic abilities

We also study the impact of logical atomic abilities on different fields. Particularly, we investigated how **conceptual understanding** supports field atomic abilities. The results in Table 6 show **significant performance improvements**, particularly in fields that rely on abstract reasoning, such as *Analysis and Geometry*. It is worth noting that activating conceptual understanding ability produces better outcomes than direct training on field-specific tasks. These results further confirm the **crucial role of conceptual understanding** as a foundational atomic ability, which highlights the need for future research to focus more on fostering **deep mathematical reasoning** and **concept-based learning**, rather than relying on **question repetition or difficulty escalation**.

5.4 Case study

To better understand the deficiencies of the model in some atomic capabilities and the impact of stimulating one atomic capability on another atomic capability, we conduct a case study. Specifically, we compare model predictions before and after stimulating algebra ability on a geometry task. As shown in Figure 3, before training, the model produces an incorrect result due to a flawed geometric assumption. After training, the model correctly applies knowledge of central angles and isosceles triangles, demonstrating the correlation from algebraic reasoning to geometric analysis.

Error Case in Topology Question: Determine the number of spanning trees of the complete graph (K_9) such that vertex 1 has degree 4. Answer: 143360 Model prediction: 8750 ✗ Model output:4 neighbors for vertex 1 (from 8 vertices), computed $\binom{8}{4} = 70$, and then assumed the rest formed a (K_5), giving $(5^3) = 125$, so $70 \times 125 = 8750$.	Error Case in Conceptual Understanding Question: Let R be a Noetherian ring. We say R is a "" if all of its local rings are complete intersections. Answer: complete intersections Model prediction: complete intersection ring ✗ Model output: a Noetherian ring (A) is called a complete intersection ring we say (A) is a complete intersection ring.
Comparison of the difference in geometric ability before and after stimulating algebraic ability Question: The following diagonal is drawn in a regular decagon, creating an octagon and a quadrilateral. What is the measure of $\angle x$? <code>import markers; \nfor(int i=0; i <=10; ++i) {\ndraw(dir(360*i/10+90)--dir(360*(i+1)/10+90));\n}\npair A = dir(360*0/10+90).....</code> Answer: 36 Model prediction before: 45 ✗ Model output:The internal angle of a regular decagon is 144° ... the angle at vertex F is half of that, so $x = 72/2 = 45$ Model prediction after: 36 ✓ Model output:The central angle between A and F is 216° ... triangle AFG is isosceles with angles at G and A equal... so angle at F is $(180-72)/2 = 36^\circ$ $(180 - 72)/2 = 36$	

Figure 3: Case study about the error cases in Topology and Conceptual Atomic ability and the comparison of stimulating one atomic capability on another atomic capability.

6 Conclusion, limitation, and future directions

In our work, we focus on decoupling the mathematical reasoning ability of LLM into atomic abilities and exploring the interactions between atomic abilities. We design field atomic capabilities and logical reasoning atomic capabilities with data. The evaluation results and analysis of decoupled atomic abilities on advanced models highlight the limited performance of some abilities. We further explore the interactions, for which we observe whether other atomic capabilities are affected when one atomic capability is stimulated. This process provides very interesting inspirations, including the facilitating role of algebraic abilities and the supporting of conceptual comprehension ability, both on logical and field atomic abilities. However, we have not explored more advanced strategies to stimulate a specific atomic capability, such as curriculum learning or reinforcement learning. Our findings will encourage exploring how to better stimulate the required atomic abilities and utilize multiple atomic abilities to solve complex mathematical tasks.

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